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(a)	Nucleus is very small compared to atom.
	The nucleus is positively charged.
	Only a small proportion of α -particles approach close to the nucleus. Electrostatic repulsion gives a large deflection.
(b)(i)	A: nucleon number = 1; proton number = 1
	B: nucleon number = 3; proton number = 1
(b)(ii)	Slow moving nuclides would not have sufficient kinetic energy to overcome large/strong electrostatic repulsion between the two nuclides.
(b)(iii)	Mass defect = 0.00240 u
	Binding energy = 3.5856×10^{-13}
	Binding energy per nucleon = $1.7928 \times 10^{-13} \text{ J}$
	= 1.1 MeV (2sf)
(c)(i)	Mean time taken for half the number of Sodium-24 present to decay / disintegrate.
(c)(ii)	$A = A_0 e^{-\lambda t}$
	$A = 1979.26187$ disintegrations per minute
	Volume of blood
	= $1979.26187 / 0.50$
	= 4000 cm^3 (2sf)
	OR
	$A = A_0 e^{-\lambda t}$
	$A_0 = 3.03143$ disintegrations per minute
	Volume of blood
	= $12000 / 3.03143$
	= 4000 cm^3 (2sf)
(c)(iii)	Short half-life of Sodium-24 makes it suitable for use as it does not stay in the body for a long period of time.
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